

The Electrochemistry & Ionic Dynamics Formula Handbook

Master Formula Index, Scaling Laws, Rapid Mental Models, and Dedicated
Micro-Numerics

Advanced Physical Chemistry Compendium

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Chapter 1

Universal Electrochemical Constants and Reference Data

Reference Constants & Critical Parameters: Electrochemical, Thermodynamic, and Transport Constants

Physical Quantity	Symbol	Standard Value (SI / Practical Units)
Faraday's Constant	F	96485.33 C/mol $e^- \approx 96500$ C/mol
Universal Gas Constant	R	8.31446 J/(mol · K) = 0.08206 L · atm/(mol · K)
Standard Temperature	T_0	298.15 K $\approx 25^\circ\text{C}$
Nernst Natural Pre-factor	RT/F	0.025693 V ≈ 25.69 mV (at 298 K)
Nernst Decimal Factor	$\frac{2.3026RT}{F}$	0.05916 V $\approx$ 0.059 V (at 298 K)
Nernst Factor at 300 K	$\frac{2.3026RT}{F}$	0.0595 V ≈ 0.060 V
Molar Gas Volume at STP	V_m	22.414 L/mol = 22400 mL/mol
H ₂ Gas STP Equivalent per Faraday	V_{H_2}	$\frac{22400}{2} =$ 11200 mL
O ₂ Gas STP Equivalent per Faraday	V_{O_2}	$\frac{22400}{4} =$ 5600 mL
Cl ₂ Gas STP Equivalent per Faraday	V_{Cl_2}	$\frac{22400}{2} =$ 11200 mL
Dielectric Constant of Water (25°C)	ϵ_r	78.54
Water Auto-ionization Constant (25°C)	K_w	$1.008 \times 10^{-14} \Rightarrow \text{p}K_w = 14.00$
Elementary Charge	e	1.60218×10^{-19} C
Avogadro's Number	N_A	6.02214×10^{23} mol ⁻¹

Dedicated Numerical Application: Rapid Charge to Faraday and Mass Conversion

Problem: A steady direct current of 4.825 A is passed through an aqueous CuSO₄ bath for 40 minutes. Calculate the moles of electrons passed and the mass of copper metal deposited ($M_{\text{Cu}} = 63.5$ g/mol).

Shortcut Calculation:

$$\begin{aligned}
 Q &= I \times t = (4.825 \text{ A})(40 \times 60 \text{ s}) = 11580 \text{ C} \\
 n_e &= \frac{Q}{F} = \frac{11580 \text{ C}}{96500 \text{ C/mol}} = \mathbf{0.120} \text{ mol } e^- \\
 n_{\text{Cu}} &= \frac{n_e}{z} = \frac{0.120}{2} = 0.060 \text{ mol} \\
 m_{\text{Cu}} &= n_{\text{Cu}} M = 0.060 \times 63.5 \text{ g} = \mathbf{3.81} \text{ g}
 \end{aligned}$$

Reference Constants & Critical Parameters: Electrochemical Series Hierarchy
(Standard Reduction Potentials at 25°C)

Reduction Half-Reaction	Couple	E° (V vs SHE)
$\text{F}_2(\text{g}) + 2\text{e}^- \longrightarrow 2\text{F}^-$	F_2 / F^-	+2.87
$\text{S}_2\text{O}_8^{2-} + 2\text{e}^- \longrightarrow 2\text{SO}_4^{2-}$	$\text{S}_2\text{O}_8^{2-} / \text{SO}_4^{2-}$	+2.01
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	$\text{MnO}_4^- / \text{Mn}^{2+}$	+1.51
$\text{Au}^{3+} + 3\text{e}^- \longrightarrow \text{Au}(\text{s})$	$\text{Au}^{3+} / \text{Au}$	+1.40
$\text{Cl}_2(\text{g}) + 2\text{e}^- \longrightarrow 2\text{Cl}^-$	$\text{Cl}_2 / \text{Cl}^-$	+1.36
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	$\text{Cr}_2\text{O}_7^{2-} / \text{Cr}^{3+}$	+1.33
$\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^- \longrightarrow 2\text{H}_2\text{O}$	$\text{O}_2 / \text{H}_2\text{O}$	+1.229
$\text{Br}_2(\text{l}) + 2\text{e}^- \longrightarrow 2\text{Br}^-$	$\text{Br}_2 / \text{Br}^-$	+1.07
$\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}(\text{s})$	Ag^+ / Ag	+0.80
$\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$	$\text{Fe}^{3+} / \text{Fe}^{2+}$	+0.77
$\text{I}_2(\text{s}) + 2\text{e}^- \longrightarrow 2\text{I}^-$	I_2 / I^-	+0.54
$\text{Cu}^{2+} + 2\text{e}^- \longrightarrow \text{Cu}(\text{s})$	$\text{Cu}^{2+} / \text{Cu}$	+0.34
$\text{Cu}^{2+} + \text{e}^- \longrightarrow \text{Cu}^+$	$\text{Cu}^{2+} / \text{Cu}^+$	+0.153
$\text{Sn}^{4+} + 2\text{e}^- \longrightarrow \text{Sn}^{2+}$	$\text{Sn}^{4+} / \text{Sn}^{2+}$	+0.15
$2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2(\text{g})$	H^+ / H_2	0.000 (Reference)
$\text{Pb}^{2+} + 2\text{e}^- \longrightarrow \text{Pb}(\text{s})$	$\text{Pb}^{2+} / \text{Pb}$	-0.126
$\text{Sn}^{2+} + 2\text{e}^- \longrightarrow \text{Sn}(\text{s})$	$\text{Sn}^{2+} / \text{Sn}$	-0.14
$\text{Ni}^{2+} + 2\text{e}^- \longrightarrow \text{Ni}(\text{s})$	$\text{Ni}^{2+} / \text{Ni}$	-0.25
$\text{Fe}^{2+} + 2\text{e}^- \longrightarrow \text{Fe}(\text{s})$	$\text{Fe}^{2+} / \text{Fe}$	-0.44
$\text{Cr}^{3+} + 3\text{e}^- \longrightarrow \text{Cr}(\text{s})$	$\text{Cr}^{3+} / \text{Cr}$	-0.74
$\text{Zn}^{2+} + 2\text{e}^- \longrightarrow \text{Zn}(\text{s})$	$\text{Zn}^{2+} / \text{Zn}$	-0.763
$2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{H}_2(\text{g}) + 2\text{OH}^-$	$\text{H}_2\text{O} / \text{H}_2, \text{OH}^-$	-0.828
$\text{Al}^{3+} + 3\text{e}^- \longrightarrow \text{Al}(\text{s})$	$\text{Al}^{3+} / \text{Al}$	-1.66
$\text{Mg}^{2+} + 2\text{e}^- \longrightarrow \text{Mg}(\text{s})$	$\text{Mg}^{2+} / \text{Mg}$	-2.37
$\text{Na}^+ + \text{e}^- \longrightarrow \text{Na}(\text{s})$	Na^+ / Na	-2.71
$\text{Ca}^{2+} + 2\text{e}^- \longrightarrow \text{Ca}(\text{s})$	$\text{Ca}^{2+} / \text{Ca}$	-2.87
$\text{K}^+ + \text{e}^- \longrightarrow \text{K}(\text{s})$	K^+ / K	-2.93
$\text{Li}^+ + \text{e}^- \longrightarrow \text{Li}(\text{s})$	Li^+ / Li	-3.04

Chapter 2

Proportionality Scaling Laws and Rapid Mental Shortcuts

Shortcut Rule / Scaling Law: Nernst Slope Scaling with Electron Transfer (n)

For any redox half-cell or full-cell reaction at 298 K, the change in cell potential per decade change ($10\times$) in the reaction quotient Q scales inversely with n :

$$\left| \frac{\Delta E}{\Delta \log_{10} Q} \right| = \frac{0.05916}{n} \text{ V} \approx \frac{59.16}{n} \text{ mV/decade} \quad (2.1)$$

Standard Decade Shifts ($Q \rightarrow 10Q$):

- For $n = 1$: Potential changes by **59.2** mV (or 0.059 V).
- For $n = 2$: Potential changes by **29.6** mV (or 0.0296 V).
- For $n = 3$: Potential changes by **19.7** mV (or 0.0197 V).
- For $n = 6$: Potential changes by **9.86** mV (or 0.0099 V).

Dedicated Numerical Application: Instantaneous Shift in Daniell Cell Potential on Dilution

Problem: In a Daniell cell $\text{Zn} | \text{Zn}^{2+}(c_1) || \text{Cu}^{2+}(c_2) | \text{Cu}$, the zinc compartment is diluted by a factor of 100 while copper concentration is held constant. By how much does the cell EMF shift?

Shortcut Calculation: The reaction quotient is $Q = [\text{Zn}^{2+}]/[\text{Cu}^{2+}]$. When $[\text{Zn}^{2+}]$ is diluted by $100\times$, Q decreases by 10^2 ($\Delta \log_{10} Q = -2$). Since $n = 2$:

$$\Delta E = -\frac{0.05916}{2} \log_{10} \left(\frac{1}{100} \right) = +0.02958 \times 2 = +\mathbf{0.0592 \text{ V}} = +\mathbf{59.2 \text{ mV}}$$

The cell EMF increases by exactly 59.2 mV.

Shortcut Rule / Scaling Law: Hydrogen Electrode Potential Scaling with pH

For the hydrogen half-cell $2\text{H}^+(\text{aq}) + 2\text{e}^- \longrightarrow \text{H}_2(\text{g}, p)$ at 298 K:

$$E_{\text{H}^+/\text{H}_2} = -0.05916 \times \text{pH} - \frac{0.05916}{2} \log_{10} p_{\text{H}_2} \quad (2.2)$$

When $p_{\text{H}_2} = 1 \text{ bar} \approx 1 \text{ atm}$:

$$E_{\text{H}^+/\text{H}_2} = -0.05916 \times \text{pH} \implies \frac{dE}{d(\text{pH})} = -59.16 \text{ mV/pH unit} \quad (2.3)$$

Key Milestones:

- At $\text{pH} = 0$ (1 M H^+): $E = 0.000 \text{ V}$.
- At $\text{pH} = 7$ (neutral water): $E = -0.05916 \times 7 = -\mathbf{0.414 \text{ V}}$.
- At $\text{pH} = 14$ (1 M OH^-): $E = -0.05916 \times 14 = -\mathbf{0.828 \text{ V}}$.

Dedicated Numerical Application: EMF of Hydrogen Concentration Cell with Differential pH

Problem: A cell consists of two hydrogen electrodes at 1 atm H_2 . The anode compartment contains gastric juice ($\text{pH} = 1.5$) and the cathode compartment contains blood serum ($\text{pH} = 7.4$). Find the cell EMF at 25°C .

Shortcut Calculation:

$$\begin{aligned} E_{\text{cell}} &= E_{\text{cathode}} - E_{\text{anode}} = -0.05916(\text{pH}_c - \text{pH}_a) \\ E_{\text{cell}} &= 0.05916 \times (\text{pH}_a - \text{pH}_c) \quad \text{Wait: Anode is oxidation! } \text{H}_2 \longrightarrow 2\text{H}^+(\text{a}) + 2\text{e}^- \\ E_{\text{cell}} &= 0.05916 \times (\text{pH}_{\text{anode}} - \text{pH}_{\text{cathode}}) \text{ is negative? No:} \\ E_{\text{cell}} &= 0.05916 \times (\text{pH}_{\text{cathode}} - \text{pH}_{\text{anode}}) = 0.05916 \times (7.4 - 1.5) \\ &= 0.05916 \times 5.9 = \mathbf{0.349 \text{ V}} \end{aligned}$$

Shortcut Rule / Scaling Law: Equilibrium Constant Shortcut from Standard EMF

At 298.15 K , the equilibrium constant K is related to E_{cell}° by:

$$\log_{10} K = \frac{nE_{\text{cell}}^\circ}{0.05916} \approx \frac{nE_{\text{cell}}^\circ}{0.0592} \approx 16.92 \times nE_{\text{cell}}^\circ \quad (2.4)$$

Mental Rules of Thumb:

- Every $+\mathbf{59.2 \text{ mV}}$ in E° yields an increase in K by a factor of $\mathbf{10^n}$.
- For $n = 1$: $E^\circ = +0.592 \text{ V} \implies K = 10^{10}$.
- For $n = 2$: $E^\circ = +0.296 \text{ V} \implies K = 10^{10}$; $E^\circ = +1.10 \text{ V} \implies K \approx 10^{37.2}$.

Dedicated Numerical Application: Equilibrium Constant of the Daniell Cell

Problem: Calculate the equilibrium constant K for $\text{Zn(s)} + \text{Cu}^{2+}(\text{aq}) \rightleftharpoons \text{Zn}^{2+}(\text{aq}) + \text{Cu(s)}$ where $E_{\text{cell}}^\circ = 1.100 \text{ V}$ at 25°C .

Shortcut Calculation: Here $n = 2$:

$$\begin{aligned} \log_{10} K &= \frac{2 \times 1.100}{0.05916} = \frac{2.200}{0.05916} = 37.187 \\ K &= 10^{0.187} \times 10^{37} = \mathbf{1.54 \times 10^{37}} \end{aligned}$$

Chapter 3

Conductance, Kohlrausch, and Transport Number Shortcuts

Shortcut Rule / Scaling Law: Fundamental Conductance Conversion Formulas

For an electrolytic solution of resistance R (Ω) and conductance $G = 1/R$ (S):

$$\text{Conductivity (Specific Conductance): } \kappa = G \left(\frac{l}{A} \right) = \frac{G^*}{R} \quad [\text{S} \cdot \text{cm}^{-1} \text{ or } \text{S} \cdot \text{m}^{-1}] \quad (3.1)$$

$$\text{Molar Conductivity: } \Lambda_m = \frac{1000\kappa}{M} \quad [\text{S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}] \quad (\kappa \text{ in S/cm, } M \text{ in mol/L}) \quad (3.2)$$

$$\text{Equivalent Conductivity: } \Lambda_{\text{eq}} = \frac{1000\kappa}{N} = \frac{\Lambda_m}{z_+} \quad [\text{S} \cdot \text{cm}^2 \cdot \text{eq}^{-1}] \quad (3.3)$$

Critical Unit Conversion Invariant:

$$\Lambda_m(\text{S} \cdot \text{m}^2 \cdot \text{mol}^{-1}) = \Lambda_m(\text{S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}) \times 10^{-4} = \frac{\kappa(\text{S} \cdot \text{m}^{-1})}{1000 \times C(\text{mol} \cdot \text{m}^{-3})} \quad (3.4)$$

Dedicated Numerical Application: Cell Constant and Molar Conductivity Calculation

Problem: A conductivity cell filled with 0.02 M KCl has resistance $R_1 = 500 \Omega$. Its conductivity is $\kappa = 0.002768 \text{ S/cm}$. When filled with 0.01 M AgNO_3 , its resistance is $R_2 = 1000 \Omega$. Find the cell constant G^* and the molar conductivity of AgNO_3 .

Shortcut Calculation:

$$\begin{aligned} G^* &= \kappa \times R_1 = 0.002768 \text{ S/cm} \times 500 \Omega = \mathbf{1.384 \text{ cm}^{-1}} \\ \kappa_{\text{AgNO}_3} &= \frac{G^*}{R_2} = \frac{1.384 \text{ cm}^{-1}}{1000 \Omega} = 1.384 \times 10^{-3} \text{ S/cm} \\ \Lambda_m &= \frac{1000 \times \kappa_{\text{AgNO}_3}}{C} = \frac{1000 \times 1.384 \times 10^{-3}}{0.01} = \mathbf{138.4 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}} \end{aligned}$$

Shortcut Rule / Scaling Law: Kohlrausch Law of Independent Migration

At infinite dilution:

$$\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ, \quad \Lambda_{\text{eq}}^\circ = \lambda_{\text{eq},+}^\circ + \lambda_{\text{eq},-}^\circ \quad (3.5)$$

Linear Subtraction Shortcut for Weak Electrolytes (HA):

$$\Lambda_m^\circ(\text{HA}) = \Lambda_m^\circ(\text{NaA}) + \Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{NaCl}) \quad (3.6)$$

For water (H_2O):

$$\Lambda_m^\circ(\text{H}_2\text{O}) = \Lambda_m^\circ(\text{HCl}) + \Lambda_m^\circ(\text{NaOH}) - \Lambda_m^\circ(\text{NaCl}) \quad (3.7)$$

Dedicated Numerical Application: Limiting Molar Conductivity of Acetic Acid

Problem: Given $\Lambda_m^\circ(\text{CH}_3\text{COONa}) = 91.0$, $\Lambda_m^\circ(\text{HCl}) = 426.0$, and $\Lambda_m^\circ(\text{NaCl}) = 126.0 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$. Determine $\Lambda_m^\circ(\text{CH}_3\text{COOH})$.

Shortcut Calculation:

$$\begin{aligned} \Lambda_m^\circ(\text{CH}_3\text{COOH}) &= \Lambda_m^\circ(\text{CH}_3\text{COONa}) + \Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{NaCl}) \\ &= 91.0 + 426.0 - 126.0 = \mathbf{391.0 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}} \end{aligned}$$

Shortcut Rule / Scaling Law: Degree of Dissociation, Ostwald Dilution, and K_{sp} Shortcuts

For a weak monobasic acid/base of concentration c :

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}, \quad K_a = \frac{c\alpha^2}{1-\alpha} \approx c\alpha^2 \implies \alpha \approx \sqrt{\frac{K_a}{c}} \quad (3.8)$$

Sparingly Soluble Salt Solubility (S) and K_{sp} Shortcut:

$$S = \frac{1000(\kappa_{\text{solution}} - \kappa_{\text{H}_2\text{O}})}{\Lambda_m^\circ(\text{salt})} \text{ mol/L} \quad (3.9)$$

Dedicated Numerical Application: Solubility and K_{sp} of BaSO_4 from Conductance

Problem: The conductivity of a saturated BaSO_4 solution at 25°C is $4.58 \times 10^{-6} \text{ S/cm}$. The conductivity of pure water is $1.52 \times 10^{-6} \text{ S/cm}$. Given $\lambda^\circ(\text{Ba}^{2+}) = 127.2$ and $\lambda^\circ(\text{SO}_4^{2-}) = 160.0 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$. Find the solubility S and K_{sp} of BaSO_4 .

Shortcut Calculation:

$$\begin{aligned} \kappa_{\text{salt}} &= (4.58 - 1.52) \times 10^{-6} = 3.06 \times 10^{-6} \text{ S/cm} \\ \Lambda_m^\circ &= 127.2 + 160.0 = 287.2 \text{ S} \cdot \text{cm}^2 \cdot \text{mol}^{-1} \\ S &= \frac{1000 \times 3.06 \times 10^{-6}}{287.2} = \frac{3.06 \times 10^{-3}}{287.2} = \mathbf{1.065 \times 10^{-5} \text{ mol/L}} \\ K_{sp} &= S^2 = (1.065 \times 10^{-5})^2 = \mathbf{1.135 \times 10^{-10} \text{ mol}^2/\text{L}^2} \end{aligned}$$

Shortcut Rule / Scaling Law: Transport Numbers and Hittorf Invariant

For cations (+) and anions (−) with ionic mobilities u_+, u_- :

$$t_+ = \frac{u_+}{u_+ + u_-} = \frac{\lambda_+^\circ}{\Lambda_m^\circ}, \quad t_- = \frac{u_-}{u_+ + u_-} = \frac{\lambda_-^\circ}{\Lambda_m^\circ}, \quad t_+ + t_- = 1 \quad (3.10)$$

Moving Boundary Method Shortcut:

$$t_+ = \frac{z_+ c V F}{Q} = \frac{z_+ c (A \cdot x) F}{I \cdot t} \quad (3.11)$$

where A is the tube cross-section, x is the boundary displacement distance, and c is concentration in mol/cm³.

Dedicated Numerical Application: Transport Number from Moving Boundary Displacement

Problem: In a moving boundary experiment with 0.01 M HCl, a current of 2.0 mA passed for 20 minutes caused the boundary to move through a distance of 8.0 cm in a tube of cross-sectional area 0.10 cm². Determine the transport number of H⁺.

Shortcut Calculation:

$$\begin{aligned} V &= A \cdot x = 0.10 \text{ cm}^2 \times 8.0 \text{ cm} = 0.80 \text{ cm}^3 = 0.80 \times 10^{-3} \text{ L} \\ n_{\text{H}^+} &= c \times V = (0.01 \text{ mol/L})(0.80 \times 10^{-3} \text{ L}) = 8.0 \times 10^{-6} \text{ mol} \\ Q &= I \cdot t = (2.0 \times 10^{-3} \text{ A})(20 \times 60 \text{ s}) = 2.40 \text{ C} \\ t_+ &= \frac{n_{\text{H}^+} \times F}{Q} = \frac{(8.0 \times 10^{-6} \text{ mol})(96500 \text{ C/mol})}{2.40 \text{ C}} = \frac{0.772}{2.40} = \mathbf{0.804} \end{aligned}$$

Chapter 4

Cell Potential, Nernst Dynamics, and Thermodynamics Shortcuts

Shortcut Rule / Scaling Law: General Multi-Electron Nernst Formula

For the general cell reaction $aA + bB \rightleftharpoons cC + dD$:

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln Q = E_{\text{cell}}^{\circ} - \frac{0.05916}{n} \log_{10} \left(\frac{a_C^c a_D^d}{a_A^a a_B^b} \right) \quad (4.1)$$

Spontaneity Criteria:

- $E_{\text{cell}} > 0 \iff \Delta G < 0$ (Galvanic / spontaneous discharge).
- $E_{\text{cell}} = 0 \iff \Delta G = 0$ (Equilibrium / completely dead cell, $Q = K$).
- $E_{\text{cell}} < 0 \iff \Delta G > 0$ (Non-spontaneous / requires external power / electrolytic).

Dedicated Numerical Application: EMF of a Galvanic Cell with Complex Stoichiometry

Problem: Calculate the EMF at 25°C for the cell:

$\text{Fe(s)} \mid \text{Fe}^{2+}(0.01 \text{ M}) \parallel \text{H}^{+}(0.1 \text{ M}) \mid \text{O}_2(0.2 \text{ atm}) \mid \text{Pt}$

given $E^{\circ}(\text{Fe}^{2+}/\text{Fe}) = -0.44 \text{ V}$ and $E^{\circ}(\text{O}_2/\text{H}_2\text{O}) = +1.23 \text{ V}$.

Shortcut Calculation:

Reaction: $2\text{Fe(s)} + \text{O}_2(\text{g}) + 4\text{H}^{+} \rightleftharpoons 2\text{Fe}^{2+} + 2\text{H}_2\text{O} \quad (n = 4)$

$$E_{\text{cell}}^{\circ} = 1.23 - (-0.44) = 1.67 \text{ V}$$

$$Q = \frac{[\text{Fe}^{2+}]^2}{p_{\text{O}_2}[\text{H}^{+}]^4} = \frac{(10^{-2})^2}{(0.2)(10^{-1})^4} = \frac{10^{-4}}{0.2 \times 10^{-4}} = \frac{1}{0.2} = 5$$

$$E_{\text{cell}} = 1.67 - \frac{0.05916}{4} \log_{10}(5) = 1.67 - 0.01479 \times 0.699 = 1.67 - 0.0103 = \mathbf{1.660 \text{ V}}$$

Shortcut Rule / Scaling Law: Cell Thermodynamics Triple Formula Matrix

From fundamental electrochemical thermodynamics:

$$\text{Free Energy: } \boxed{\Delta G = -nFE_{\text{cell}}, \quad \Delta G^\circ = -nFE_{\text{cell}}^\circ} \quad (4.2)$$

$$\text{Entropy: } \boxed{\Delta S = nF \left(\frac{\partial E}{\partial T} \right)_P} \quad (4.3)$$

$$\text{Enthalpy: } \boxed{\Delta H = \Delta G + T\Delta S = -nFE + nFT \left(\frac{\partial E}{\partial T} \right)_P = -nF \left[E - T \left(\frac{\partial E}{\partial T} \right)_P \right]} \quad (4.4)$$

Sign of Temperature Coefficient $(\partial E/\partial T)_P$:

- If $\left(\frac{\partial E}{\partial T} \right)_P > 0$: $\Delta S > 0$, cell absorbs heat from surroundings during reversible discharge ($q_{\text{rev}} > 0$).
- If $\left(\frac{\partial E}{\partial T} \right)_P < 0$: $\Delta S < 0$, cell releases heat to surroundings during reversible discharge ($q_{\text{rev}} < 0$).
- If $\left(\frac{\partial E}{\partial T} \right)_P = 0$: $\Delta H = \Delta G = -nFE$, cell EMF is temperature invariant.

Dedicated Numerical Application: Enthalpy and Entropy from Temperature Coefficient

Problem: A reversible cell has $E = 1.018$ V at 298 K and temperature coefficient $\left(\frac{\partial E}{\partial T} \right)_P = -5.0 \times 10^{-5}$ V/K with $n = 2$. Find ΔG , ΔS , and ΔH at 298 K.

Shortcut Calculation:

$$\Delta G = -2 \times 96500 \times 1.018 = -\mathbf{196.47} \text{ kJ/mol}$$

$$\Delta S = 2 \times 96500 \times (-5.0 \times 10^{-5}) = -\mathbf{9.65} \text{ J/(mol} \cdot \text{K)}$$

$$\Delta H = \Delta G + T\Delta S = -196474 + 298(-9.65) = -196474 - 2876 = -\mathbf{199.35} \text{ kJ/mol}$$

Chapter 5

Concentration Cells and Liquid Junction Potentials

Shortcut Rule / Scaling Law: Electrode-Concentration vs Electrolyte-Concentration Cells

1. Electrode Concentration Cell (Amalgams / Gas Pressure):

$$\text{H}_2(p_1) | \text{H}^+(c) | \text{H}_2(p_2) \implies E = \frac{RT}{2F} \ln \left(\frac{p_1}{p_2} \right) = \frac{0.05916}{2} \log_{10} \left(\frac{p_1}{p_2} \right) \quad (5.1)$$

Spontaneous when $p_1 > p_2$.

2. Electrolyte Concentration Cell without Transference (Salt Bridge Present):

$$\text{M} | \text{M}^{z+}(c_1) || \text{M}^{z+}(c_2) | \text{M} \implies E = \frac{0.05916}{z} \log_{10} \left(\frac{c_2}{c_1} \right) \quad (5.2)$$

Spontaneous discharge requires $c_2 > c_1$ (cathode concentrated, anode dilute).

3. Electrolyte Concentration Cell WITH Transference (No Salt Bridge): For a 1 : 1 univalent electrolyte (HCl or AgNO₃):

$$E_{\text{with transference}} = 2t_- \frac{RT}{F} \ln \left(\frac{c_2}{c_1} \right) = 2t_- \frac{0.05916}{1} \log_{10} \left(\frac{c_2}{c_1} \right) \quad (5.3)$$

Liquid Junction Potential (E_j) Master Shortcut:

$$E_j = E_{\text{with transference}} - E_{\text{without transference}} = (2t_- - 1) \frac{RT}{F} \ln \left(\frac{c_2}{c_1} \right) = (t_- - t_+) \frac{0.05916}{z} \log_{10} \left(\frac{c_2}{c_1} \right) \quad (5.4)$$

When $t_+ = t_- = 0.5$ (e.g., saturated KCl salt bridge where $t_{\text{K}^+} \approx t_{\text{Cl}^-} \approx 0.49$), $E_j \rightarrow 0$.

Dedicated Numerical Application: Liquid Junction Potential Calculation

Problem: A concentration cell with transference has 0.01 M and 0.10 M HCl solutions in direct contact. Given $t_{\text{H}^+} = 0.82$ and $t_{\text{Cl}^-} = 0.18$. Calculate E_{without} , E_{with} , and the liquid junction potential E_j at 25°C.

Shortcut Calculation:

$$E_{\text{without}} = 0.05916 \log_{10} \left(\frac{0.10}{0.01} \right) = 0.05916 \times 1 = \mathbf{0.0592 \text{ V}}$$

$$E_{\text{with}} = 2t_{-}(0.05916) \log_{10}(10) = 2 \times 0.18 \times 0.05916 = \mathbf{0.0213 \text{ V}}$$

$$E_j = E_{\text{with}} - E_{\text{without}} = 0.0213 - 0.0592 = \mathbf{-0.0379 \text{ V} = -37.9 \text{ mV}}$$

Chapter 6

Faraday's Laws, Gas Volumes, and Overpotential Shortcuts

Shortcut Rule / Scaling Law: Faraday's Unified Electrolysis Solver

For any substance X undergoing an n -electron redox process:

$$m = \frac{M \cdot I \cdot t \cdot \eta}{n \cdot F}, \quad n_X = \frac{I \cdot t \cdot \eta}{n \cdot F} \quad (6.1)$$

where $\eta = \frac{\text{actual mass}}{\text{theoretical mass}}$ is the current efficiency.

Gas Volume at STP Rapid Mental Conversion: For 1 Faraday (96500 C) of charge:

$$2 \text{H}^+ + 2 \text{e}^- \longrightarrow \text{H}_2 \implies V_{\text{H}_2} = \frac{22400}{2} = \mathbf{11200} \text{ mL} \quad (6.2)$$

$$2 \text{H}_2\text{O} \longrightarrow \text{O}_2 + 4 \text{H}^+ + 4 \text{e}^- \implies V_{\text{O}_2} = \frac{22400}{4} = \mathbf{5600} \text{ mL} \quad (6.3)$$

$$2 \text{Cl}^- \longrightarrow \text{Cl}_2 + 2 \text{e}^- \implies V_{\text{Cl}_2} = \frac{22400}{2} = \mathbf{11200} \text{ mL} \quad (6.4)$$

Combined Water Electrolysis Gas Ratio:

$$V_{\text{total gas}}(\text{H}_2 + \text{O}_2) = 11200 + 5600 = \mathbf{16800} \text{ mL/Faraday at STP} \quad (6.5)$$

Dedicated Numerical Application: Total Gas Evolved During Water Electrolysis

Problem: Acidulated water is electrolyzed with a constant current of 0.50 A for 1930 seconds. Calculate the total volume of wet gas collected at STP ($\text{H}_2 + \text{O}_2$).

Shortcut Calculation:

$$Q = 0.50 \text{ A} \times 1930 \text{ s} = 965 \text{ C}$$

$$\text{Number of Faradays } f = \frac{965 \text{ C}}{96500 \text{ C/F}} = 0.010 \text{ F}$$

$$V_{\text{total}} = f \times 16800 \text{ mL} = 0.010 \times 16800 = \mathbf{168} \text{ mL at STP}$$

$$(V_{\text{H}_2} = 112 \text{ mL}, \quad V_{\text{O}_2} = 56 \text{ mL})$$

Shortcut Rule / Scaling Law: Electrode Overpotential and Butler-Volmer Tafel Shortcut

Actual Deposition Potential:

$$E_{\text{actual}} = E_{\text{reversible}} + \eta_{\text{overpotential}} \quad (6.6)$$

High-Overpotential Tafel Equation: At high anodic/cathodic polarization ($|\eta| > 0.1$ V):

$$\eta = a + b \log_{10} I = \frac{2.303RT}{\alpha nF} \log_{10} \left(\frac{I}{I_0} \right) \quad (6.7)$$

The Tafel slope $b \approx \frac{59.2}{\alpha n}$ mV/decade at 25°C.

Dedicated Numerical Application: Current Shift from Overpotential Shift

Problem: An electrochemical cathode reaction follows the Tafel relationship with slope $b = 0.118$ V/decade. By what factor does the net current density increase if the cathodic overpotential η is increased from 0.20 V to 0.436 V?

Shortcut Calculation:

$$\Delta\eta = 0.436 - 0.200 = 0.236 \text{ V}$$

$$\Delta \log_{10} I = \frac{\Delta\eta}{b} = \frac{0.236 \text{ V}}{0.118 \text{ V/decade}} = 2.0$$

$$\frac{I_2}{I_1} = 10^{\Delta \log_{10} I} = 10^2 = \mathbf{100\times}$$

Chapter 7

Batteries, Fuel Cells, and Corrosion Kinetics

Shortcut Rule / Scaling Law: Commercial Cell Energy Density and Capacity Shortcuts

$$\text{Theoretical Capacity: } C_{\text{th}} = \frac{nF}{M} \quad [\text{C/g or A} \cdot \text{h/g}] \quad (1 \text{ A} \cdot \text{h} = 3600 \text{ C}) \quad (7.1)$$

$$\text{Theoretical Specific Energy: } \mathcal{E} = C_{\text{th}} \times E_{\text{cell}} \quad [\text{W} \cdot \text{h/kg}] \quad (7.2)$$

$$\text{Fuel Cell Thermodynamic Efficiency: } \eta_{\text{th}} = \frac{\Delta G^\circ}{\Delta H^\circ} = \frac{-nFE^\circ}{\Delta H^\circ} \quad (7.3)$$

H₂–O₂ Fuel Cell Benchmarks at 298 K: $\Delta G^\circ = -237.2 \text{ kJ/mol}$, $\Delta H^\circ = -285.8 \text{ kJ/mol}$, $n = 2$:

$$E^\circ = \frac{237200}{2 \times 96485} = 1.229 \text{ V}, \quad \eta_{\text{th}} = \frac{-237.2}{-285.8} = \mathbf{83.0\%} \quad (7.4)$$

Dedicated Numerical Application: Theoretical Specific Capacity of Lithium Anode

Problem: Calculate the theoretical specific capacity of pure metallic lithium ($M = 6.94 \text{ g/mol}$, $z = 1$) in $\text{A} \cdot \text{h/g}$ and $\text{mA} \cdot \text{h/g}$, and compare it with zinc ($M = 65.4 \text{ g/mol}$, $z = 2$).

Shortcut Calculation:

$$C_{\text{th}}(\text{Li}) = \frac{1 \times 96500 \text{ C/mol}}{6.94 \text{ g/mol} \times 3600 \text{ s/h}} = \frac{96500}{24984} = \mathbf{3.863 \text{ A} \cdot \text{h/g}} = \mathbf{3863 \text{ mA} \cdot \text{h/g}}$$

$$C_{\text{th}}(\text{Zn}) = \frac{2 \times 96500 \text{ C/mol}}{65.4 \text{ g/mol} \times 3600 \text{ s/h}} = \frac{193000}{235440} = \mathbf{0.820 \text{ A} \cdot \text{h/g}} = \mathbf{820 \text{ mA} \cdot \text{h/g}}$$

$$\text{Capacity ratio } \frac{\text{Li}}{\text{Zn}} = \frac{3863}{820} \approx \mathbf{4.71 \times}$$

Shortcut Rule / Scaling Law: Corrosion Rate and Penetration Thickness Shortcut

By Faraday's law, a uniform corrosion current density i_{corr} (A/cm^2) corrodes a metal surface at a penetration depth rate:

$$\text{Corrosion Rate (CR)} = \frac{i_{\text{corr}} \cdot M}{z \cdot F \cdot \rho} \quad [\text{cm/s}] \quad (7.5)$$

In practical engineering units (mm/year):

$$\text{CR (mm/year)} = 3.27 \times 10^{-3} \times \frac{i_{\text{corr}}(\mu\text{A/cm}^2) \cdot M(\text{g/mol})}{z \cdot \rho(\text{g/cm}^3)} \quad (7.6)$$

Dedicated Numerical Application: Annual Steel Tank Corrosion Depth

Problem: A steel storage tank ($M_{\text{Fe}} = 55.85 \text{ g/mol}$, $z = 2$, $\rho = 7.87 \text{ g/cm}^3$) is exposed to sea water. Electrochemical polarization measurements reveal a corrosion current density of $i_{\text{corr}} = 15.0 \mu\text{A/cm}^2$. What thickness of steel is lost per year?

Shortcut Calculation:

$$\begin{aligned} \text{CR} &= 3.27 \times 10^{-3} \times \frac{15.0 \times 55.85}{2 \times 7.87} \\ &= 3.27 \times 10^{-3} \times \frac{837.75}{15.74} = 3.27 \times 10^{-3} \times 53.22 = \mathbf{0.174 \text{ mm/year}} \end{aligned}$$